# Spanning Equiangular Intervals for Basso's Relative Projection Constants

Run `fa_banach_001`, human-requested LLM follow-up

2026-06-15

## Source Problem and Review Prompt

This note follows up on the reviewed packet `1901.07866_pi_7_3_equiangular_lines`. The source is Giuliano Basso's *Computation of maximal projection constants*, arXiv:1901.07866 [1]. Basso asks in Remark 4.6 whether there are integers $n \leq d$ such that

$$\Pi(n, d) = \frac{7}{3}.$$

The earlier packet answered this by proving $\Pi(7, 15) = 7/3$. The human review found the argument correct but recommended pushing beyond the isolated value by looking for a systematic pattern or an infinite family.

as claimed.

**Remark 4.6** As $\mathbb{R}^5$ admits a system of ten equiangular lines, cf. [LS73, p. 496], the general upper bound of König, Lewis, and Lin, cf. [KLL83], tells us that

$$\Pi(5, 10) = 2.$$

To summarize, we obtain the sequence

$$\Pi(1,1) = 1, \quad \Pi(2,3) = \frac{4}{3}, \quad \Pi(4,6) = \frac{5}{3}, \quad \Pi(5,10) = 2,$$

which naturally leads to the open question: Are there integers $n \leqslant d$ such that

$$\Pi(n, d) = \frac{7}{3}?$$

**4.5 Acknowledgements:** I am thankful to Urs Lang who read earlier draft versions of this article and who stimulated several improvements. Moreover, I am grateful to Anna Bot for proofreading this paper.

Figure 1: Basso, Remark 4.6, where the 7/3 question is posed.

## KLL Criterion Used

Basso recalls the theorem of Koenig–Lewis–Lin [3]:

$$\Pi(n,d) \le B(n,d) := \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}\left(1-\frac{n}{d}\right)}.$$

The equality case is used here in the full-dimensional form also used in the human review of the original packet: equality is certified by $d$ distinct equiangular lines whose representatives span an $n$-dimensional Euclidean space. This full-dimensional condition is essential. A lower-dimensional equiangular configuration embedded into a larger $\mathbb{R}^n$ is not, by itself, a KLL equality certificate for $\Pi(n,d)$.

**1.3 Main result** Our main result reads as follows.

**Theorem 1.2** *If* $n \geqslant 1$ *is an integer, then*

$$\Pi_n = \sup_{d \geqslant 1} \max \left\{ \frac{n}{d} + \frac{1}{d}\sum_{k=1}^{n} \lambda_k(T) \,:\, T \textit{ is a } K_{n+2}\textit{-free two-graph of order } d \right\}.$$

To prove Theorem 1.2, we invoke a simple trick, see Lemma 2.2, that allows us to greatly narrow down the matrices that need to be considered. This is done in Section 2.

**1.4 Relative projection constants** The following question has first been systematically addressed by König, Lewis, and Lin in [KLL83]:

**Question 1.1** *Let* $n, d \geqslant 0$ *be integers. What is*

$$\Pi(n,d) := \sup\left\{\Pi(E) : E \subset \ell_d^\infty \textit{ is an } n\textit{-dimensional Banach space}\right\}?$$

By definition, $\sup \varnothing = -\infty$. Clearly, $\Pi(d,d) = 1$ and it is a direct consequence of the classical Hahn-Banach theorem that $\Pi(1,d) = 1$ for all integers $d \geqslant 1$. The quantity $\Pi(d-1,d)$ has been examined by Bohnenblust, cf. [Boh38], where it is shown that $\Pi(d-1,d) \leqslant 2 - \frac{2}{d}$. In [CL09], Chalmers and Lewicki determined the exact value of $\Pi(3,5)$. In [KLL83], König, Lewis, and Lin established the general upper bound

$$\Pi(n,d) \leqslant \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}\left(1-\frac{n}{d}\right)}$$

with equality if and only if $\mathbb{R}^n$ admits a system of $d$ distinct equiangular lines. Thereby, as $\mathbb{R}^3$ admits a system of six equiangular lines, cf. [LS73, p. 496], it holds that

$$\Pi(3,6) = \frac{1+\sqrt{5}}{2}.$$

In light of

$$\Pi(4,6) = \frac{5}{3},$$

which we demonstrate in Paragraph 4.4, up to $d = 6$ all exact values of $\Pi(n,d)$ for $1 \leqslant n \leqslant d$ are now computed. It is well-known that

Figure 2: Basso's definition of $\Pi(n,d)$ and the quoted Koenig–Lewis–Lin equality criterion.

## Main Outcome

The corrected systematic principle is the spanning interval principle: if $M$ equiangular lines span $\mathbb{R}^n$, then one gets exact KLL values for every number $d$ of lines between $n$ and $M$. This turns the original $\Pi(7, 15) = 7/3$ observation into the full interval

$$\Pi(7, d) = B(7, d) \qquad (7 \le d \le 28),$$

and Paley conference matrices give infinitely many further intervals

$$\Pi\left(\frac{q+1}{2}, d\right) = B\left(\frac{q+1}{2}, d\right) \qquad \left(\frac{q+1}{2} \le d \le q+1,\ q \equiv 1 \pmod 4\right).$$

For Basso's specific value 7/3, the KLL upper bound itself reaches 7/3 at exactly two integer pairs:

$$(n, d) = (7, 15), \qquad (n, d) = (49, 51).$$

Both are sharp. The first is supplied by the $E_7$ 28-line configuration. The second is supplied by the Paley conference matrix for $q = 97$, which gives 98 equiangular lines spanning $\mathbb{R}^{49}$; a 51-line spanning subsystem certifies

$$\Pi(49, 51) = \frac{7}{3}.$$

## Why the Spanning Condition Matters

It is tempting to use equiangular systems in lower-dimensional subspaces and then embed them into larger Euclidean spaces. This would be too strong. For example, six equiangular lines in $\mathbb{R}^3$ can be embedded into $\mathbb{R}^4$, but Basso records

$$\Pi(4, 6) = \frac{5}{3},$$

whereas the KLL upper bound $B(4, 6)$ is larger than 5/3. Thus a mere lower-dimensional embedding cannot be the intended equality condition. All equality certificates below are therefore full-dimensional: the selected lines span the relevant $\mathbb{R}^n$.

## Proof Intuition

The previous proof used one 15-line subset of a classical 28-line system in dimension 7. Since seven lines in that chosen system already form a basis, one can add any further lines from the same equiangular configuration and keep a spanning equiangular system. That gives every $d$ from 7 to 28 at once.

The same mechanism works for conference matrices. A symmetric conference matrix of order $2n$ yields $2n$ equiangular lines spanning $\mathbb{R}^n$, so every $d$ between $n$ and $2n$ is KLL-exact. Paley's finite-field construction gives infinitely many such conference matrices. Taking $q = 97$ gives exactly the new 7/3 pair $(49, 51)$.

## The Spanning Interval Principle

**Theorem 1** (Spanning interval principle)**.** *Suppose $\mathbb{R}^n$ contains $M$ distinct equiangular lines spanning $\mathbb{R}^n$. Then, for every integer $d$ with $n \le d \le M$,*

$$\Pi(n,d) = \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}\left(1-\frac{n}{d}\right)}.$$

*Proof.* Choose nonzero representatives of the $M$ lines. Since the representatives span $\mathbb{R}^n$, some $n$ of them are linearly independent. Fix those $n$ basis lines and add any $d-n$ further lines from the original equiangular system. The selected $d$ lines remain equiangular and still span $\mathbb{R}^n$. The Koenig–Lewis–Lin equality criterion gives the displayed equality. □

## The $E_7$ Interval

**Corollary 1.** *For every integer $d$ with $7 \le d \le 28$,*

$$\Pi(7,d) = \frac{7}{d} + \sqrt{(d-1)\frac{7}{d}\left(1-\frac{7}{d}\right)}.$$

*In particular,* $\Pi(7,15) = 7/3$.

*Proof.* Let

$$H = \left\{ x \in \mathbb{R}^8 : \sum_{i=1}^{8} x_i = 0 \right\}.$$

For each two-element subset $A \subset \{1,\ldots,8\}$, define

$$v_A = \sum_{i\in A} e_i - \frac{1}{4}(1,\ldots,1) \in H.$$

For two such sets $A, B$,

$$\langle v_A, v_B \rangle = |A \cap B| - \frac{1}{2}.$$

Thus the 28 lines $\mathbb{R}v_A$ are equiangular in $H \cong \mathbb{R}^7$. The seven vectors $v_{\{1,j\}}$, $2 \le j \le 8$, form a basis of $H$: if

$$\sum_{j=2}^{8} a_j v_{\{1,j\}} = 0$$

and $A = \sum_{j=2}^{8} a_j$, then the coordinates $2,\ldots,8$ give $a_j = A/4$ for every $j$, hence $A = 7A/4$, so all $a_j = 0$. The spanning interval principle applies. □

## Conference and Paley Intervals

**Theorem 2** (Conference intervals)**.** *Suppose there is a symmetric conference matrix of order $2n$. Then, for every integer $d$ with $n \le d \le 2n$,*

$$\Pi(n,d) = \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}\left(1-\frac{n}{d}\right)}.$$

*Proof.* Let $C$ be a symmetric conference matrix of order $2n$:

$$C_{ii} = 0, \qquad C_{ij} \in \{\pm 1\} \ (i \neq j), \qquad C^2 = (2n-1)I.$$

Since $C$ is symmetric, its eigenvalues are $\pm\sqrt{2n-1}$. Also $\operatorname{tr} C = 0$, so the two eigenspaces have the same dimension, namely $n$. The projection

$$P = \frac{1}{2}\left(I + \frac{1}{\sqrt{2n-1}}C\right)$$

has rank $n$, diagonal entries $1/2$, and off-diagonal entries of common absolute value $1/(2\sqrt{2n-1})$. Hence $P$ is the Gram matrix of $2n$ equiangular lines spanning $\mathbb{R}^n$. The spanning interval principle applies. $\square$

**Corollary 2** (Paley intervals)**.** *Let $q$ be an odd prime power with $q \equiv 1 \pmod 4$, and put*

$$n = \frac{q+1}{2}.$$

*Then, for every integer $d$ with $n \leq d \leq q+1$,*

$$\Pi(n,d) = \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}\left(1 - \frac{n}{d}\right)}.$$

*At the endpoint,*

$$\Pi\left(\frac{q+1}{2}, q+1\right) = \frac{1+\sqrt{q}}{2}.$$

*Proof.* Let $\mathbb{F}_q$ be the finite field with $q$ elements, and let $\chi : \mathbb{F}_q \to \{0, \pm 1\}$ be the quadratic character. Index rows and columns by $\mathbb{F}_q \cup \{\infty\}$, and define

$$C_{\infty,\infty} = 0, \qquad C_{\infty,x} = C_{x,\infty} = 1 \quad (x \in \mathbb{F}_q),$$

$$C_{x,x} = 0, \qquad C_{x,y} = \chi(x-y) \quad (x, y \in \mathbb{F}_q,\ x \neq y).$$

Since $q \equiv 1 \pmod 4$, $\chi(-1) = 1$, so $C$ is symmetric. The standard quadratic-character identity

$$\sum_{t \in \mathbb{F}_q} \chi((t-a)(t-b)) = -1 \qquad (a \neq b)$$

implies $C^2 = qI_{q+1}$. Hence $C$ is a symmetric conference matrix of order $q+1 = 2n$. The conference interval theorem applies. $\square$

## The KLL Route to $7/3$

**Lemma 1** (Fixed target arithmetic)**.** *Fix a rational number $c > 1$. The integer pairs $n \leq d$ satisfying*

$$B(n,d) = c$$

*are finite and are given by the formula*

$$d = \frac{n(n+1-2c)}{n-c^2} = n + \frac{n(c-1)^2}{n-c^2},$$

*whenever the right-hand side is an integer at least $n$.*

*Proof.* The equation $B(n,d)=c$ is equivalent to

$$\sqrt{(d-1)\frac{n}{d}\left(1-\frac{n}{d}\right)}=c-\frac{n}{d}.$$

Squaring and simplifying gives the displayed formula. Since $d\geq n$, the second expression forces $n>c^2$. For $n>c^2$, the difference

$$d-n=\frac{n(c-1)^2}{n-c^2}$$

is a decreasing function of $n$ and is bounded. Hence only finitely many integer values of $m=d-n$ can occur. For each such $m$, the equation

$$m(n-c^2)=n(c-1)^2$$

determines at most one value of $n$. Therefore only finitely many integer pairs can occur. $\square$

**Theorem 3.** *The integer pairs $n\leq d$ for which the Koenig–Lewis–Lin upper bound $B(n,d)$ is exactly $7/3$ are precisely*

$$(n,d)=(7,15)\quad\textit{and}\quad(n,d)=(49,51).$$

*For both pairs the bound is sharp. Therefore*

$$\Pi(7,15)=\Pi(49,51)=\frac{7}{3}.$$

*Proof.* For $c=7/3$, the fixed-target formula gives

$$d=n+\frac{16n}{9n-49}.$$

As $d\geq n$ and $d\neq n$, put $m=d-n\geq 1$. Then

$$(9m-16)n=49m.$$

The denominator must be positive, so $m\geq 2$, and the condition $n\geq 6$ forces $m\leq 19$. Also $9m-16$ divides $49m$, hence it divides

$$9(49m)-49(9m-16)=784.$$

Among positive divisors $r$ of 784 with $r=9m-16$ and $2\leq m\leq 19$, only $r=2$ and $r=56$ occur. These give

$$(m,n,d)=(2,49,51)\quad\text{and}\quad(8,7,15).$$

The pair $(7,15)$ is sharp by the $E_7$ interval. For $(49,51)$, use the Paley construction with $q=97$. Since 97 is prime and $97\equiv 1 \pmod 4$, it gives 98 equiangular lines spanning $\mathbb{R}^{49}$. The spanning interval principle gives

$$\Pi(49,51)=\frac{49}{51}+\sqrt{50\cdot\frac{49}{51}\cdot\frac{2}{51}}=\frac{49}{51}+\frac{70}{51}=\frac{7}{3}.$$

$\square$

**Remark 1.** *This theorem classifies the pairs for which the KLL upper bound itself equals $7/3$. It is not a full classification of all conceivable pairs satisfying $\Pi(n,d)=7/3$, because a pair with $B(n,d)>7/3$ could in principle have exact value $7/3$ by some non-KLL mechanism.*

## Large Classical Intervals

The same spanning interval principle turns every known full-dimensional equiangular-line system into exact relative projection constants. In particular, the classical Gerzon-sharp systems in dimensions $2, 3, 7, 23$ give

$$\Pi(2,d) = B(2,d) \qquad (2 \leq d \leq 3),$$
$$\Pi(3,d) = B(3,d) \qquad (3 \leq d \leq 6),$$
$$\Pi(7,d) = B(7,d) \qquad (7 \leq d \leq 28),$$

and, using the classical 276 equiangular lines in $\mathbb{R}^{23}$,

$$\Pi(23,d) = B(23,d) \qquad (23 \leq d \leq 276).$$

The last interval is included as a literature-backed application rather than as a new construction in this note; the 276-line system is standard in the equiangular-lines literature [4, 7].

## A Small "Thirds Ladder"

The fixed-target arithmetic can be applied to neighboring values $k/3$. The table below lists selected KLL-bound candidates that are certified by the spanning equiangular systems used above. It is not a classification table for these other values; its role is to show that Basso's 7/3 question is part of a broader exact-value mechanism.

| value | certified pairs $(n,d)$ | source |
|---|---|---|
| 4/3 | $(2,3)$ | simplex |
| 5/3 | $(5,6)$ | simplex or Paley $q = 9$ |
| 2 | $(5,10)$ | Paley $q = 9$ |
| 7/3 | $(7,15), (49,51)$ | $E_7$; Paley $q = 97$ |
| 3 | $(13,26), (15,25), (21,28)$ | Paley $q = 25, 29, 41$ |
| 3 | $(27,33), (45,50)$ | Paley $q = 53, 89$ |

## Literature Exploration and Further Tools

The original source has a later erratum [2]. The erratum concerns an error in Lemma 3.1 and a different part of the computation of maximal projection constants. The present note uses the KLL/equiangular-line criterion quoted in Section 1.4 of the original paper, not the affected lemma.

The equiangular-line side has much more raw material than is used above. Lemmens–Seidel [4] is the classical reference for equiangular lines, conference matrices, and the low-dimensional exceptional systems. Greaves–Koolen–Munemasa–Szollosi [6] develop Seidel-matrix and computational tools for equiangular lines in Euclidean spaces. Lin–Yu [7] gives modern structural context around the Lemmens–Seidel conjecture and the special Gerzon-sharp dimensions. Fixed angle asymptotics, as in Jiang–Tidor–Yao–Zhang–Zhao [8], suggest a possible source of broad lower bounds, though exact KLL equality still requires actual full-dimensional finite equiangular systems. Other equiangular tight frame constructions, such as the Tremain and Steiner/Hadamard families studied by Fickus–Jasper–Mixon–Peterson [9], are promising future inputs for more exact $\Pi(n,d)$ intervals.

For a full 7/3 classification, the main missing ingredient would be a way to rule out non-KLL mechanisms for pairs whose KLL upper bound is larger than 7/3. Basso's two-graph/eigenvalue framework and linear-programming formulations of relative projection constants look like the relevant tools, but they are outside the short equiangular argument of this note.

## Verification Notes

The script `code/verify_systematic_families.py` checks the finite certificates and arithmetic used in this packet.

First, it verifies with exact rational arithmetic that the $E_7$ model gives 28 equiangular lines spanning $\mathbb{R}^7$, and that the interval $7 \le d \le 28$ can be selected so as to remain spanning. It also checks that the $d = 15$ KLL value is exactly $7/3$.

Second, it verifies the integer search behind the KLL-bound equation $B(n, d) = 7/3$, recovering only $(7, 15)$ and $(49, 51)$.

Third, it constructs Paley conference matrices for all primes $q \equiv 1 \pmod 4$ up to 100 and verifies $C^2 = qI$. For $q = 97$, it also checks with exact arithmetic over $\mathbb{Q}(\sqrt{97})$ that the first 51 Paley lines have Gram rank 49, giving an explicit spanning certificate for $\Pi(49, 51) = 7/3$.

Finally, it verifies the sample thirds-ladder entries displayed above.

## Limitations and Novelty

The packet does not claim a new equiangular-line construction. Its contribution is the translation: known full-dimensional equiangular-line systems give intervals of exact relative projection constants. The most concrete payoff for Basso's question is

$$\Pi(7, 15) = \Pi(49, 51) = \frac{7}{3},$$

together with a proof that these are the only KLL-bound equality points at the value $7/3$.

The result stops short of a complete classification of $\Pi(n, d) = 7/3$. Such a classification would need control over possible non-KLL equality mechanisms when the KLL upper bound is strictly larger than $7/3$.

## Human Review Recommendation

Review should focus on four points. First, confirm the exact full-dimensional form of the Koenig–Lewis–Lin equality criterion. Second, check the spanning interval principle. Third, check the arithmetic classification of KLL-bound equality at $7/3$. Fourth, check the Paley $q = 97$ spanning certificate for the new pair $(49, 51)$.

## References


[1] G. Basso, *Computation of maximal projection constants*, arXiv:1901.07866, 2019. https://arxiv.org/abs/1901.07866

[2] G. Basso, *Erratum to "Computation of maximal projection constants"*, arXiv:2402.06672, 2024. https://arxiv.org/abs/2402.06672

[3] H. Koenig, D. R. Lewis, and P.-K. Lin, *Finite dimensional projection constants*, Studia Math. 75 (1983), 341–358.

[4] P. W. H. Lemmens and J. J. Seidel, *Equiangular lines*, J. Algebra 24 (1973), 494–512.

[5] R. E. A. C. Paley, *On orthogonal matrices*, J. Math. Phys. 12 (1933), 311–320.

[6] G. R. W. Greaves, J. H. Koolen, A. Munemasa, and F. Szollosi, *Equiangular lines in Euclidean spaces*, arXiv:1403.2155, 2014. https://arxiv.org/abs/1403.2155

[7] Y.-T. Lin and W.-H. Yu, *Equiangular lines and the Lemmens-Seidel conjecture*, arXiv:1807.06249, 2018. https://arxiv.org/abs/1807.06249

[8] Z. Jiang, J. Tidor, Y. Yao, S. Zhang, and Y. Zhao, *Equiangular lines with a fixed angle*, Ann. of Math. 194 (2021), 729–743. https://arxiv.org/abs/1907.12466

[9] M. Fickus, J. Jasper, D. G. Mixon, and J. Peterson, *Tremain equiangular tight frames*, arXiv:1602.03490, 2016. https://arxiv.org/abs/1602.03490